SNMP

At the end of this episode, I will be able to:

1. Describe the basics about the Simple Network Management Protocol (SNMP)

Learner Objective: *Learn about the Simple Network Management Protocol (SNMP)*

Description: In this episode, you will learn about the Simple Network Management Protocol (SNMP) used in many networks today. This discussion includes information about the functions and operation of this protocol as well as the versions found in use today.

- Here are some of the main components of SNMP:

- Management Information Base (MIB) - the Management Information Base (MIB) in SNMP serves as a structured database that organizes and stores information about network devices and their parameters for monitoring and management purposes.
- The SNMP Agent - the SNMP agent functions as a software module on a network device, facilitating communication between the device and the SNMP manager by collecting and providing access to management information via the Simple Network Management Protocol.
- Traps - traps in SNMP are asynchronous notifications sent by a network device to an SNMP manager, alerting it of specific events or conditions, enabling proactive monitoring and response to changes in the network.

- There are currently two major versions of SNMP that can be found in use in networks today.
- SNMP v2c was the popular version for decades. A major issue with this version is its lack of security. Security is based on a clear text password called a community string.
- SNMP v3 is the latest version of SNMP and addresses security head on. This version offers built-in security for both encryption and authentication. In fact, the protocol goes a step further and offers multiple levels of both encryption and authentication as well as multiple combinations.

Additional Resources:

What is SNMP and How Does it Work?
https://www.fortra.com/resources/articles/snmp-basics-what-it-and-how-it-works